VERSION 1.0.0 · ENGLISH EDITION

Complete Organizer Manual

Setup, operation, judging, reporting and known limits

University Contest Platform · PC² 9.10.0 / WTI 1.1

Omar Al-Mukhtar University - Libya

Dashboard development and customization: Boamtoal

First-release manual, including documented source revisions through 23 October 2026.

Original PC²/WTI developers and ICPC identity retain their rights. See attribution and operating conditions. No real team credentials or private competition material is included.

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Dashboard workspace tree

One connected system tree. Tool names match the actual interface. Owner marks restricted advanced tools. The structure is generated from the real navigation definition.

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Contest preparation and operation

1. Prepare host and network

Install the matching PC²/WTI runtime, isolated judge and router/LAN. Keep the clock paused.

↓

2. Rehearse independently

Use synthetic problems and the hidden diagnostic team. Test every physical laptop and recovery.

↓

3. Prepare official draft

Set teams, problems/tests, resources, limits, colors, schedule, penalty, freeze and judging mode.

↓

4. Activate services

Only one managed event is active. Activation does not start the contest clock.

↓

5. Start and supervise

Start explicitly. Review pending judgements, clarifications, clocks and connection incidents.

↓

6. Finalize and retain

Resolve pending work, release freeze deliberately, save reports/backups and share per-team feedback privately.

Submission, judging and results

1. Team submits source

Problem + deployed language + main source file. Receipt reduces duplicate retries.

↓

2. Pending / isolated execution

Compilation, execution and validation occur inside the deployed judge boundary.

↓

3. Provisional automatic verdict

Automatic only may finalize; review mode holds the result for a human. Human only skips automatic execution.

↓

4. One human confirmation

Approve the displayed exact verdict, or choose a different precise one. Owner access and audit remain required.

↓

5. Final result and standings

Notify the team. Only final accepted solutions score; public freeze hides unreleased outcomes.

↓

6. Post-contest learning

Capture retained evidence and a separately labelled current-data replay. Never rewrite official results for the report.

CHAPTER 01

Start here: what this platform does

> User Guide → Start here

University Contest Platform v1.0.0 adds an organizer dashboard, participant interface customization and operational tools around PC² 9.10.0 and its Web Team Interface (WTI). PC² remains the contest engine and the authority for native submissions and official judgements.

The dashboard was developed and customized by Boamtoal in the context of programming contests at Omar Al-Mukhtar University (OMU), Libya. It helps organizers prepare and operate contests; it does not replace or claim authorship of PC². Original PC² and WTI credits, logos and license notices must remain intact.

Read Installation before a fresh deployment, then rehearse the entire event on the intended router and team laptops. Version 1 is an initial open-source release. Successful local checks are not certification for every host, university network or contest size.

After the updated interface has loaded, refreshing the browser restores the signed-in dashboard account's section, contextual tab and available page position. Saved navigation is scoped to that account and revalidated against its current role. Blocked browser storage falls back safely to Overview.

Steps

1. Install the supported runtime and isolated judge on a dedicated organizer computer.
2. Create a separate rehearsal contest and test the complete participant journey.
3. Prepare a new official contest with fresh team credentials and reviewed problem data.
4. Take a recoverable backup, verify the clock is paused, and start it explicitly at the intended time.

CHAPTER 02

Network, router and participant addresses

> System → Participant test & access

Use a real router or dedicated access point, preferably with the organizer server wired to it. A phone hotspot can exhaust DHCP leases, disconnect clients or isolate devices. In the original event, moving from the phone hotspot to a router resolved the observed connectivity problem. This is an operational recommendation, not proof that every router is configured correctly.

The organizer and teams must share a reachable local network. Disable access-point/client isolation on the contest LAN and provide enough DHCP leases. Keep the organizer address stable with a router DHCP reservation when possible. Internet access is not required for local contest traffic after dependencies and the judge VM have been installed.

The sidebar shows the detected server IP and last update. Use the exact participant URL from the access center, including scheme and port. The standard LAN entry is http://SERVER-IP:8088/#/login when HTTP has been explicitly enabled. 127.0.0.1 is only the computer that opens it; never send the dashboard loopback URL to teams.

Automatic address detection can update the displayed address and managed listener after a network change. It cannot rewrite a URL already typed on another laptop. Teams must open the new address. A short hostname works only when the network actually resolves it; an IP address is the reliable fallback.

A fully wired Ethernet contest works without Wi-Fi. Connect the organizer and team computers by Ethernet to the same router or switch. A router normally supplies DHCP; a switch alone needs an existing DHCP service or correctly assigned static addresses. A USB-to-Ethernet adapter can provide a port for the organizer laptop. Use the wired interface’s actual address in participant links.

Steps

1. Before the event, open the login page on every physical team laptop.
2. If switching networks during an event, pause the contest explicitly first and communicate the interruption.
3. Move the server and every team to the same router, read the new participant address, and have teams reopen it.
4. Verify existing submissions and pending review, then resume the clock explicitly. Do not create a new contest to fix Wi-Fi.

> **Important limit:** Browser heartbeat monitoring is not a router device inventory or proof of physical network presence. A local rehearsal on the server cannot certify access from other laptops.

CHAPTER 03

Sign-in, roles and account separation

> My Account; Security & Logs

The private local dashboard is http://127.0.0.1:8091/ on the server itself. A first owner account named admin receives a random temporary password in the selected state folder at admin-dashboard/INITIAL-LOGIN.txt. Read it locally, change it immediately, and keep it away from teams. The temporary file is removed after a successful password change.

Dashboard passwords must contain 16–128 characters. Enable TOTP two-step verification in My Account; the enrollment QR is generated locally. Store the enrollment secret securely. Password changes invalidate that user’s sessions.

Viewer is read-only. Operator can perform permitted contest and resource operations. Owner also controls dashboard security, sensitive native administration, manual verdict decisions and rejudging. Some advanced native tools are owner-only even if a related basic form is available to an operator.

Dashboard users are separate from native PC² teams, judges and service accounts. Creating a dashboard operator does not create a native judge, and creating a judge account does not launch a judge process. Never give a team the organizer credentials or reuse the internal bridge account in the native Administrator.

Steps

1. Use an individual named dashboard account for each organizer.
2. Give each person the least privilege needed.
3. Review sessions and revoke access when an organizer leaves.

CHAPTER 04

Optional access for another organizer

> System → Remote dashboard access

Local-only administration is the default. Optional remote administration uses a separate HTTPS listener on an explicitly selected private LAN address, normally port 8092. A dashboard account and an unexpired account-specific IP approval are both required; approving an address alone does not sign anyone in.

Choose the user, verify the actual source IP, and approve it for a limited period (1–168 hours). Inspect the displayed HTTPS URL and certificate instructions on the remote laptop. DHCP changes can require approval of the new address. Revoke the approval and the user's sessions when access is no longer needed.

Participant access settings do not expose the organizer dashboard through the team proxy. Do not forward organizer ports on a public router, bypass TLS checks casually, or expose this initial local deployment as a public Internet service.

> **Important limit:** This release is not a certified public cloud or multi-tenant hosting platform. Account/IP approval complements authentication; it cannot protect a compromised organizer computer.

CHAPTER 05

Create a contest and keep history separate

> Contests

Use a new contest for a new event. A draft has its own identity, clock settings, accounts, resources and later native state. Select a clear title and short name, unique lowercase ID, duration, penalty and judging workflow. Choose explicitly whether to copy problems and resources from the selected template.

New official team credentials must be generated for the new event. Historical runs are not copied as new attempts. Each new contest has a reserved private diagnostic account. Prepare drafts while another contest is active, but only one native contest server may be active through the managed lifecycle.

Selecting a draft changes the workspace you edit; it does not move participant traffic or start its clock. Activation is a separate operation, and clock start is separate again. A second-active-contest warning prevents concurrent managed activation.

Deactivate only after pausing the clock and resolving unfinished submissions. Activation failures can leave a reserved stage rather than a safely completed deployment. Use Recover server for the matching reservation; a failure before secure bootstrap requires operator inspection. Do not delete journals or overwrite an initialized native profile to force it to work.

Steps

1. Create and name the draft.
2. Add official teams, reviewed problem data, statements and the intended schedule.
3. Rehearse on an independent technical contest.
4. Back up the current contest and safely deactivate it if necessary.
5. Activate the new contest, verify service health and keep the clock paused until the event starts.

CHAPTER 06

Problems, difficulty and balloon colors

> Problems → Problems; Edit settings

A problem has a stable native ID, letter, title, short name, active flag, limits, validator profile, judge data and optional resources. Edit it from its own row. The difficulty column is organizer metadata; an estimated difficulty is labelled as estimated and does not affect scoring.

Balloon color belongs to the problem. Save both the color name and color value; details show the name and swatch. Colors identify problems on scoreboards, with accessible letters/tooltips retained. Balloon service settings control delivery or printing, not problem color.

Adding a judged problem requires an owner and a paused contest. Choose a deployed isolated language/validator profile and upload paired private tests. New problems start inactive; check their limits, data and published statements before enabling them.

The current policy accepts a time limit of 1–10 seconds and memory limit of 32–1024 MiB. A test ZIP may contain up to 250 matching .in/.ans pairs, at most 20 MiB compressed and 50 MiB expanded. Larger datasets require an intentionally reviewed import/deployment path, not bypassing validation.

> **Important limit:** Changing a title, test set, difficulty or resource does not rejudge older attempts automatically. Current stored tests may differ from those used by a historical submission.

CHAPTER 07

Private tests and trusted execution

> Problems → Private judge data; Judging → Judge clients

Each test pairs an input file with its expected answer, for example 01.in and 01.ans. Review the problem statement, reference solution, limits and validator behavior together. An incorrect expected answer can produce an incorrect automatic decision even when the software runs correctly.

Compilation and execution of untrusted team code belong to the dedicated isolated Linux judge, not the organizer host. The deployed policy uses an isolated Lima VM, no shared host directories, restricted execution, resource limits and a separate broker. Do not enable host compilers for real untrusted submissions.

The currently deployed isolated profiles cover C++17, C++20 and Python. Custom executables, arbitrary compiler/validator uploads, additional languages and host architectures need trusted engineering and independent validation. Saving a custom language inactive does not make it safe or supported.

Private judge inputs, expected answers, reference solutions and organizer source are not participant resource uploads. Keep them private during the contest; post-contest learning exports intentionally disclose selected material.

> **Important limit:** The isolation implementation has local regression checks, not an independent security certification. No software-only claim eliminates all risk from hostile code or a privileged host attacker.

CHAPTER 08

PDF statements, Arabic versions and supporting files

> Resources

Attach each file to the correct problem, or choose a contest-wide booklet. Use the resource type and language to distinguish an English statement, Arabic statement, bilingual PDF and supporting material. Statements and booklets must be PDF; supported general attachments include PDF, TXT, CSV, JSON, ZIP, PNG and JPG.

Uploads are drafts until explicitly published. Participants see only published resources linked to active problems after signing in. Publishing a replacement statement supersedes the same problem/type/language version. Earlier versions remain available privately. Unpublishing blocks direct aliases as well as catalog visibility.

A file may be at most 20 MiB. Retained resources have a 1 GiB quota, including removed versions. Removed files can be restored as drafts. Archives are treated as data, never as runnable installers.

New PDF uploads reject common active/embedded features; export a plain PDF if rejected. These checks are not antivirus scanning. Check every intended participant download and language label before the event.

Steps

1. Upload reviewed statements and supporting files.
2. Preview metadata and publish deliberately.
3. Sign in as the diagnostic team and verify access.
4. Confirm that a signed-out browser cannot access problem resources.

CHAPTER 09

University identity, logos and colors

> System → Dashboard identity; Participant identity

Dashboard identity controls the organizer sidebar name, image and identity colors, including active/hover/focus states. Participant identity separately controls the contestant experience, main labels, logos and participant theme. Do not assume one theme automatically follows the other.

Keep readable contrast. Semantic colors for Accepted, Pending, errors, destructive actions and balloon identities are not ordinary branding colors. Changing institutional identity must not make a failed verdict look successful.

The default identity reflects Omar Al-Mukhtar University and its LCPC context. Another university can configure its own identity. Preserve original PC²/WTI attribution and logos in About, bundled notices and upstream interfaces; university customization does not transfer ownership of PC².

Participant pages and new dashboard UI are English. The standalone participant Instructions page is the deliberate English/Arabic exception. A problem statement may independently be Arabic, English or bilingual.

CHAPTER 10

Official teams, credentials and the private test team

> Teams & Accounts; System → Participant test & access

Create or import team accounts, set display names, login access and public standings visibility. Account permissions are checkboxes in native tools. Credential generation downloads are private; distribute each team only its own login details. Avoid posting credential lists in chat, public repositories or screenshots.

The reserved team99999 is the organizer diagnostic team. It uses the same participant routes, clock restrictions and judging workflow, but is hidden from official standings, official counts and balloon delivery. Its activity is retained separately for diagnosis. Do not edit or overwrite this reserved identity.

An owner can download test access after reauthentication and configured TOTP. Use that login in a separate browser profile or private window, ideally on a second physical laptop. A separate rehearsal contest is safer for full submission experiments before an official event.

The diagnostic account helps reproduce participant behavior; it does not prove every team’s Wi-Fi, browser, firewall or DHCP configuration works. Before the actual contest, have every official device open the intended URL and sign in.

CHAPTER 11

Schedule, start, pause and remaining time

> Contest Controls → Clock & schedule

Choose a country and IANA time zone, then use the calendar/time picker. Review both the local time and UTC preview. Changing zone converts the same saved instant; editing the calendar changes the intended instant. Saving a schedule or activating a server does not itself press Start.

At the event, explicitly start the contest from Contest Controls. Sensitive actions require a reason or typed CONFIRM where shown. Pause for an operational interruption before changing networks or performing maintenance. Adding time changes the contest duration; verify the remaining clock afterward.

The remaining-time display advances locally from the server sample and becomes red when 15% of the original duration remains. A stale connection is labelled rather than being presented as a trusted fresh clock. Check the server's actual state before acting on an offline display.

Freeze remaining time is a scoreboard setting, not a pause of judging or of the clock. Stopping services is different from pausing the contest. A browser refresh, opening a report or signing out does not pause the native event.

> **Important limit:** Rehearse the intended scheduled behavior and time zone on the installed PC² version. Do not infer that saving a date guarantees an unattended event start.

CHAPTER 12

Participant submissions and result dialogs

> Submissions

Teams select a problem, supported language and source file in the participant interface. A received attempt stays Pending while queued, executing or waiting for human approval. Final Accepted is green; Wrong Answer, Compilation Error, Runtime Error and other unsuccessful final results are red.

Final result notifications can appear across participant pages. They report the server’s final decision, not a guessed answer from raw yes/no flags. If the connection drops while sending, check My submissions before trying again. An uncertain receipt is not permission to blindly resubmit.

Submission receipts reduce duplicates for retries with the same retained client key; they are not automatic recovery of all browser state. A full reload may require file reselection. Browser download completion means handoff to the browser, not a guarantee that the file was saved to disk.

CHAPTER 13

Automatic judging and human approval

> Judging → Pending human review; Submissions → Awaiting human

Choose Automatic only, Automatic + human approval, or Human only. Change the workflow while paused and after finalizing pending attempts. Automatic modes require a running judge service. Human only queues manual decisions without automatic execution.

In Automatic + human approval, the automatic verdict is provisional. It remains pending for teams and does not count in standings until an authorized owner confirms a final decision. Open a submission row to see its automatic result, source and retained evidence.

Approve result confirms the exact displayed automatic verdict, including Wrong Answer or a runtime/compile failure. A single confirmation shows the attempt, team, problem and final verdict; no typed reason or CONFIRM is required for this pending approval. An automatic reason is retained in the audit log. Choose different verdict opens one selection/confirmation when an override is needed. Reviewing an already final decision or rejudging retains the separate deliberate review flow.

Manual changes and rejudges affect official results and are audited. Check the current record after a timeout before retrying; native PC² operations are not a distributed database transaction. A new judgement can coexist with retained older evidence, which remains labelled separately.

Steps

1. Open the pending queue.
2. Read the automatic result, code and relevant saved evidence.
3. Confirm the displayed automatic verdict, or choose a different precise verdict if needed.
4. Confirm once, then verify the final state and participant notification.

CHAPTER 14

Read source, output and test-case evidence

> Click a submission row → Details

Click a row for a concise summary; editing remains an explicit button. The Details action opens an organized page for source, current private tests and retained compiler/execution/validator evidence. Viewing never compiles, submits or rejudges code.

Separate Input, actual Team output and Expected output. A validator’s decision is authoritative for its configured matching rules; a visual whitespace difference alone is not a new judgement. Green Passed and red Failed reflect retained native test flags. Not validated and Missing are distinct states; no test passes are invented.

PC² may stop at the first failed test. Saved stdout/stderr without a reliable case number is shown as an unassigned report; the dashboard does not guess which input produced it. Compile/runtime error headers are red and relevant failure streams open automatically.

At most ten retained native reports and 64 KiB per text preview are displayed in this view. Empty and missing files differ. Per-case execution time is shown when retained; peak memory and historic test-data revisions are not available in this integration.

CHAPTER 15

Private live and public frozen scoreboards

> Scoreboard & Awards → Live Scoreboard / Public Scoreboard

Live Scoreboard is a private organizer view that ignores the public freeze. Public Scoreboard follows the participant freeze and is the view intended for audience display. Never project the private live screen during a freeze.

Problem columns use assigned balloon colors with letter/name context. Cells reflect native scoring, attempts, pending state and accepted time. Pending human review does not become a solved problem. Color alone is not the only problem identifier.

Configure penalties, scoring, freeze and finalization deliberately. Thawing releases the frozen public results; it is different from ending the clock. Review unresolved judgements and final ranking before releasing or certifying results.

Balloon delivery and award tools depend on the corresponding native configuration. Email, printers and external drivers are deployment prerequisites; enabling a setting does not create working printer/mail infrastructure.

A 1ST crown badge and a darker accepted-green cell identify First to solve for each problem. It uses original elapsed submission milliseconds among final accepted, visible official teams, not the order of human approval. Exact-time ties are shared. Deleted/rejudged results recalculate the marker. The public view reveals only first solves whose accepted cell has been released by the native freeze policy; the marker does not change rank or penalty.

Balloon delivery lists one entry per final accepted team/problem. Click Mark delivered, review the team, problem and balloon color, then Confirm delivery. No typed reason or CONFIRM is needed. Cancel leaves the entry unchanged. Mark pending reverses the acknowledgement through the same simple confirmation. This records delivery status; it does not claim a balloon was physically handed over or that a printer/email service worked.

CHAPTER 16

Questions and announcements

> Clarifications

Teams send clarification requests through the participant interface. Open the relevant row, read the question and choose whether an answer is private or sent to all teams. Review the audience before confirming.

Use announcements for operational notices, corrected statements and remaining-time messages. Do not disclose private tests, credentials or another team’s source. If a resource changes, publish its reviewed replacement and explicitly explain the change to the intended teams.

CHAPTER 17

Team activity, connection warnings and notifications

> Teams & Accounts → Participant activity / Team connections

The activity timeline stores bounded diagnostic metadata, request failures and verified interactions. It does not record passwords, full source files or unrestricted browser content. Reported team identities are distinguished from identities verified by the participant session.

Connection monitoring aggregates browser tabs. A roughly three-minute heartbeat gap or a closed-session grace interval can produce a connection event. A hidden tab, sleeping laptop, network interruption or browser closure may all affect this signal. It is not a scan of the router’s connected devices.

The notification bell near Local control channel retains security alerts, participant incidents and loss/restoration events. Temporary toasts disappear; stored notifications remain for review. Repeated events can be grouped. Read the event’s timestamp and detail before assuming a new incident.

The diagnostic history is bounded (currently 30 days / 200,000 events). Its HMAC chain helps detect alteration within the retained records; a privileged attacker who controls the host and keys can defeat local guarantees. Connection durations are observed browser-session durations, not proof a device continuously stayed on one physical network.

Steps

1. Filter the list by team, time and event category near the records.
2. Check whether a failure is isolated or affects several teams.
3. Correlate request errors with router and server health.
4. Export only the necessary private diagnostic data and follow institutional retention rules.

CHAPTER 18

CSV, PDF and recovery backups

> Exports & Backups

CSV/PDF reports summarize contest information. They are not a restorable backup. Some native reports contain credentials or private settings and require owner reauthentication. Keep all organizer exports private until their contents have been reviewed.

A full encrypted backup includes the retained contest and dashboard material selected by the backup workflow. Take it with the required services cleanly stopped for consistency. Keep the encryption/recovery key separately; losing the key can make recovery impossible.

Restore creates a new workspace rather than silently overwriting the current contest. Test recovery on a separate copy before relying on it for an event. A contest archive is not a complete disk image of macOS, the installed native dependencies or the judge VM.

Downloads show a spinner during preparation and actual received-byte progress when a reliable Content-Length exists. Unknown lengths use an indeterminate indicator and byte count. Do not interpret an estimated server work percentage as final disk-save completion.

CHAPTER 19

Detailed post-contest learning packages

> Exports & Backups → Team Learning Reports

An owner can prepare learning packages after the contest is finished and paused, or its native server is stopped. Only teams with retained submissions, their own clarifications or verified interactive activity qualify. Passive heartbeats, unverified identities and the hidden diagnostic team are excluded.

Each participating team receives its own folder and ZIP containing a detailed PDF, report.json, retained source, attempted problem statements, organizer reference code/reasoning, and available evidence. The master archive contains all independent team folders; give a team only its own archive.

Upload reference solutions and reasoning as UTF-8 source or Markdown/text (up to 512 KiB per file), linked to the correct stable problem ID. Difficulty and estimated status, attempt counts, chronology, official decisions and available organizer reasons are included. A textual source comparison is not proof of correctness.

A separately labelled isolated replay executes supported team code against current stored tests. Each replayed case has Input, actual Team output and Expected output. Full raw files accompany marked long previews. C++ reference code uses a monospaced presentation; the original source is retained unchanged.

Historical official judgements remain unchanged. Current-data replay is not a reconstruction of an unavailable historic dataset. Saved native output without a reliable test association remains separate from new replay output. Missing evidence and execution failures are explicitly labelled.

Preparation runs in the background with completed-work progress; a downloadable package appears only after all required team PDFs succeed. The local export folder is admin-dashboard/learning-reports/<export-id>/ beneath that contest’s private state.

> **Important limit:** These archives intentionally contain solutions and private judge data. Release them only after the contest and after checking reuse rights, privacy and future problem reuse. Replay currently supports the deployed C++17/C++20/Python profiles, not arbitrary native languages.

CHAPTER 20

Security, privacy and operating conditions

> Security & Logs

Use the local dashboard on a trusted, updated organizer machine. Keep OS access controlled, use individual strong credentials, enable TOTP, review account-specific remote approvals and protect backups. Do not publish .local, runtime logs, databases, secrets or participant source.

The HTTP team listener is intended only for an explicitly approved private LAN. HTTP does not encrypt credentials or source in transit. Use the supported HTTPS path with correctly trusted certificates where traffic confidentiality is required; a private address alone is not encryption.

Audit events record actions and metadata with chain verification. Browser diagnostics are a separate bounded stream. Inform participants what is collected, who may inspect it, how long it is retained and how feedback archives are shared, following the university's applicable policies.

Owners must have permission to use problem statements, reference code, logos and student submissions. Code licensing does not grant trademark rights or permission to republish private contest material. Maintain original notices and attribute contributions when redistributing the source.

Version 1 is provided under the repository license without a promise of uninterrupted operation, perfect scoring inputs or complete protection from compromise. The organizer remains responsible for rehearsal, data correctness, final judgements, lawful use and recovery planning.

CHAPTER 21

Troubleshooting during an event

> System; Team connections; Security & Logs

Login page does not open: verify the exact scheme, server IP and port, then the same reachable LAN, DHCP lease and router client isolation. A Windows 169.254.x.x address with no gateway usually indicates no usable DHCP lease. Reconnect to the intended router and obtain a valid address; an Internet connection is not required merely to obtain a local DHCP lease.

On the affected Windows laptop, run ipconfig, then ping SERVER-IP and PowerShell Test-NetConnection SERVER-IP -Port 8088 (use the configured port). Ping failure alone is inconclusive because ICMP may be blocked; the TCP port test is more relevant to the web page. Do not assign a random static address without knowing the subnet, gateway and an unused address.

Participant Request failed is an HTTP/network symptom, not automatically a wrong password or wrong solution. Inspect the request/event details and service health. If many teams fail together, check the router, server listener and DHCP before changing account credentials.

Pending submissions: distinguish queued execution from Awaiting human. Check judge health and the human review queue. Never approve results merely to clear a technical blockage. A timeout may follow a successful native operation; inspect the actual record before retrying.

Dashboard reconnecting: verify its selected state and native services. Guide and About remain readable after dashboard sign-in even without a live contest snapshot. A signed-in page can hold stale data; watch its last update and connection banner.

No retained per-test output: inspect available evidence and its labels. Do not invent actual output from an expected answer. Use a post-contest isolated replay if appropriate, clearly distinguished from the original judgement.

Rejected PDF upload: export a plain non-active PDF and retry within limits. Failed PDF export: check the configured Python runtime and reportlab/pypdf dependencies. Download spinner without a percentage can mean the server does not know the response length.

CHAPTER 22

Native Administrator coverage and boundaries

> System → Supported features

The dashboard organizes native tools under their relevant sections: problems/languages/tests; accounts/groups/imports; submission management/comparison; judging; scoring/balloons; clocks; profiles/playback/reset; reports; sites/connections. Sensitive tools require an owner, confirmation and sometimes reauthentication.

The native Java Administrator remains available under a separate account and retains its original About/credits. Use the coverage matrix to see supported operations. This dashboard does not claim complete replacement of every Swing dialog, plugin or deployment mode.

The installed upstream Event Feed add/update methods are unimplemented. Use supported native JSON/XML reports instead. Shadow CCS status and Java plugin inventory are read-only here; the dashboard does not install or execute arbitrary plugins.

Playback requires a separate empty technical contest and explicit start. Native playback can start clocks and continues independently of the browser; stop it through its controls. Reset is destructive, requires a full saved backup and exact confirmation, and is not the normal way to prepare a new event.

Bulk native operations may partially complete if a connection fails. Inspect actual state before retrying. Printers, mail, external validators and multi-site native deployments require their own tested setup.

CHAPTER 23

Version 1 limitations and contribution priorities

> About; GitHub → Issues / Pull requests

The validated organizer deployment is macOS on Apple Silicon with a dedicated Lima VZ ARM64 Linux judge. Native PC² itself supports more systems, but this overlay's host lifecycle and isolation are not certified on Windows, Linux organizer hosts or Intel Macs.

Team and remote-administrator browser clients are separate from the organizer host. They may access the supported server from Windows, macOS or Linux browsers; rehearse the actual browsers and LAN. A Linux guest judge does not establish Linux organizer-host support, and Java-based PC² does not automatically port this overlay.

There is no one-click, fully validated clean-machine installer. Install the matching upstream PC²/WTI distribution, build the bridge, configure tool paths and provision the isolated judge according to Installation. Clean-machine installation/restore and the intended physical load still require a rehearsal.

Historic dataset revisions, trustworthy per-test association for every retained native output and peak memory measurements are unavailable. Reports label those gaps. Replay and PDF generation can be expensive for many attempts/tests; capacity planning and more incremental export work remain useful improvements.

Independent security auditing, broader OS/architecture support, external-service integration and larger event load certification remain open contribution areas. See ROADMAP for a maintained list. Do not advertise unfinished features as completed.

Everyone may study, use, modify and contribute under EPL-2.0. Open an issue with sanitized reproduction steps, or a pull request with focused changes and relevant tests. Never attach credentials, real student code, private tests or university network inventories to public issues.

CHAPTER 24

Before, during and after the contest

> Organizer checklist

Before: rehearse installation and backup recovery; use a router; open the participant page on every laptop; verify accounts, problem limits, validators, statements, reference solutions, colors, time zone, duration, penalty, freeze and human-review staffing. Check the hidden test account and confirm the clock is paused.

During: start explicitly, monitor judge health and pending review, review notifications, answer clarifications, check last-update timestamps and keep private standings off the public display. Pause before a planned network or service interruption.

After: finalize pending decisions, verify rankings and thaw intentionally. Save reports and an encrypted recovery backup. Prepare learning archives only after permission to disclose solutions/tests, distribute per-team archives privately and apply the university’s retention policy.

Appendix A: Installation and first rehearsal

This is a **source overlay**, not a self-contained replacement distribution of PC². The procedure below is for an experienced organizer on the documented host. It has not been validated as a one-click clean-machine installer. Do a complete rehearsal and recovery test before scheduling a real event.

Supported baseline

Operating-system compatibility

Machine / role	Current status	What this means
Organizer server: macOS on Apple Silicon	Current exercised deployment	Follow this guide, then rehearse a clean installation, recovery and the intended client load.
Organizer server: Intel Mac	Not supported by this overlay release	ARM64 judge images, VZ settings and host paths require porting and validation.
Organizer server: Windows	Not supported by this overlay release	Service lifecycle, networking and isolated judge provisioning need a Windows-specific implementation.
Organizer server: Linux	Not supported by this overlay release	Replace macOS/Lima host integration with a reviewed Linux deployment and isolated execution boundary.
Team laptops: Windows, macOS or Linux	Browser clients	No local Java/PC² installation is required for the web interface. Rehearse the actual browsers, file uploads and LAN access on every laptop.
Optional remote organizer browser	Browser client with approved access	It connects to the supported server over the configured HTTPS path; it does not move server services to the remote laptop.

PC² being Java-based does not make this custom overlay portable automatically. The Linux judge is a guest execution environment, not validation of Linux as an organizer host. Host scripts currently depend on macOS tools, filesystem paths, Lima VZ and ARM64 compiler/runtime profiles. Do not run team code directly on another organizer host as a shortcut around the missing isolation deployment.

For a research-based starting point, read Other operating systems: experimental porting notes or the Arabic edition. Those suggestions cite official sources and **have not been tested with this platform**. They do not change the support status above.

Host prerequisites

• Dedicated **macOS Apple Silicon** organizer host with enough RAM/storage for Java, PDF generation and a **4 GiB RAM / 20 GiB disk** Lima VZ Linux ARM64 judge. Keep at least 8 GiB free host storage; large learning exports need significantly more space.

• Matching official **PC² 9.10.0 build 7065** distribution, including `lib/pc2.jar`, its dependency jars and `projects/WebTeamInterface-1.1-7065.zip`. Obtain it from official PC² distribution information. Verify the distribution's provenance/checksum and preserve its original licenses, manuals and About credits.

• Java capable of building with `--release 17` (Java 17 is the bridge build baseline). The deployed native runtime also uses `-Djdk.crypto.KeyAgreement.legacyKDF=true` for its storage compatibility. New JDK/upstream combinations need independent validation.

• Python **3.11+**, an isolated virtual environment with PDF requirements, Node **20+** for JavaScript tests, Lima, OpenSSL supporting CMS AES-GCM, Caddy, and macOS command-line tools (`xcrun`, `lsof`, `ssh`).

• Optional `clang-format` for presentation-only C++ formatting in learning PDFs. Original source remains intact if presentation formatting is unavailable.

• A real router/dedicated AP, enough DHCP leases, no team-client isolation and preferably a wired organizer host. Initial dependency/VM downloads require Internet; local event traffic afterward does not.

The bundled VM template pins a specific Ubuntu ARM64 image and SHA-256. If it is unavailable, review and deliberately update **both** URL and digest; do not disable image verification. Guest package repositories require access during initial provisioning.

1. Install the source overlay beside the native distribution

Work in a **fresh copy** of the official PC² installation. Keep any existing event installation separate and backed up. Clone this repository into a separate directory, then copy the source overlay into the fresh native root:

```
git clone https://github.com/Boamtoal/pc2-university-platform.git
cp -R pc2-university-platform/lcpc /path/to/fresh-pc2-9.10.0/
cp -R pc2-university-platform/tools /path/to/fresh-pc2-9.10.0/
cp -R pc2-university-platform/docs /path/to/fresh-pc2-9.10.0/
```

Do not replace the upstream `README`, `VERSION`, `doc`, `bin`, `lib` or About/license assets. Keep this repository's documentation and NOTICE available to operators too. Commands below run from `/path/to/fresh-pc2-9.10.0`; replace that placeholder with your real directory.

2. Python and build dependencies

```
cd /path/to/fresh-pc2-9.10.0
python3 -m venv .venv
. .venv/bin/activate
python -m pip install -r lcpc/admin/requirements.txt
mkdir -p work/wti .local/tools
unzip -q projects/WebTeamInterface-1.1-7065.zip -d work/wti
export LCPC_BUILD_JAVA_HOME="$JAVA_HOME"
sh lcpc/build-tools.sh
```

Set `JAVA_HOME` to an installed JDK first, or set `LCPC_BUILD_JAVA_HOME` directly. The expected extracted build dependency is `work/wti/WebTeamInterface-1.1/WebTeamInterface-1.1.jar`. The source package excludes event-specific import/audit programs; the reusable bridge sources remain included.

Install Lima, Caddy and a suitable OpenSSL through your trusted package manager. Copy the trusted Caddy executable to `.local/tools/caddy` (the current lifecycle expects that path) and make it executable. For example, after installation:

```
cp "$(command -v caddy)" .local/tools/caddy
chmod 700 .local/tools/caddy
```

Do not download an arbitrary executable from an issue/comment. Configure runtime paths explicitly if your installation differs:

Variable	Meaning
`LCPC_JAVA_HOME` / `JAVA_HOME`	Native Java home
`LCPC_JAVA_BIN`	Java executable if no Java home is selected
`LCPC_BUILD_JAVA_HOME`	JDK for compiling the bridge
`LCPC_LIMA_BIN`	Trusted `limactl` executable
`LCPC_OPENSSL`	OpenSSL executable for the participant network manager
`LCPC_PDF_PYTHON`	Python executable with reportlab/pypdf; defaults to the running Python
`LCPC_STATE_DIR`	Selected private native contest state for `pc2ctl.py`

Some older lifecycle helpers still expect Homebrew OpenSSL at `/opt/homebrew/bin/openssl`, and macOS tools at their standard locations. See Roadmap before attempting another host layout. Start the dashboard with the same virtual-environment Python or explicitly configure `LCPC_PDF_PYTHON`.

3. Provision the isolated judge

Read `lcpc/infra/judge-vm.yaml` and `lcpc/infra/linux/` before provisioning. The VM has **no host directory mounts**. Never add your contest or home directory as a shared mount to make an error disappear.

```
export LIMA_HOME="$PWD/.local/lima"
limactl start --tty=false --name=lcpc-judge lcpc/infra/judge-vm.yaml
python -B lcpc/pc2ctl.py linux-install
```

Initial VM provisioning installs Linux packages and the trusted sandbox runtime; it does not initialize an official contest or start its clock. Re-running setup against a modified judge requires operator review. An existing incompatible VM must not be silently replaced.

4. Bootstrap a fresh technical rehearsal

Do this only in the fresh installation with no other managed server using the native ports. The supplied practice template contains **two public synthetic toy problems**, no real event tests or solutions.

```
python -B lcpc/pc2ctl.py init-isolated-test rehearsal
export LCPC_STATE_DIR="$PWD/.local/instances/rehearsal"
python -B lcpc/pc2ctl.py bootstrap
python -B lcpc/pc2ctl.py isolated-up
python -B lcpc/admin/manage.py init --state "$LCPC_STATE_DIR"
python -B lcpc/admin/manage.py start --state "$LCPC_STATE_DIR"
```

Bootstrap is only for a new uninitialized instance. It generates fresh private credentials and secures native accounts. Never rerun it over an initialized event. A partial failure must be inspected rather than deleting profiles to force a restart. Startup leaves the contest clock paused.

Open **http://127.0.0.1:8091/** locally. Read the random owner password from `$LCPC_STATE_DIR/admin-dashboard/INITIAL-LOGIN.txt`, sign in as `admin`, replace it with a unique 16–128 character password and enable TOTP. Do not include this file in any public archive. The native Administrator, when required, uses a separate account.

5. Configure private participant access

Read the access center and network policy first. Prefer HTTPS for confidentiality. The supported TLS path creates a local CA that clients must intentionally trust; do not train participants to ignore certificate warnings.

For an explicitly approved plaintext private LAN rehearsal, identify the organizer's actual IP and subnet first, then use the network manager:

```
python -B lcpc/pc2ctl.py network-http-prepare SERVER-IP LAN-CIDR
python -B lcpc/pc2ctl.py network-confirm SERVER-IP LAN-CIDR
python -B lcpc/pc2ctl.py network-trial
```

`SERVER-IP` and `LAN-CIDR` are placeholders, not valid literal arguments. Use only a verified private network you administer. For TLS use `network-prepare SERVER-IP LAN-CIDR` and follow the generated CA instructions before `network-trial`. Default participant HTTP is port **8088**, TLS **8443**. The dashboard is **8091** loopback; optional remote organizer HTTPS is a separate **8092** listener with user/IP approval.

Check **System → Participant test & access** for the actual URLs and read-only checks. The UI's adaptive IP display does not change addresses already typed on other laptops. Test on the real router and every physical team device.

6. Rehearse and prepare the official event

Use the separate diagnostic team in a separate browser profile, verify statements/downloads, send known successful and failing synthetic solutions, inspect exact verdicts and human approval, and compare private/public freeze behavior. Starting the **technical** clock for these tests is an explicit operator action. Never test by changing official event results.

Create the official event through **Contests**, using a new unique ID and fresh teams. Review problem data, references, limits, validators, resources, difficulty, colors, duration, timezone, penalties, approval mode and freeze. Safely deactivate the rehearsal before activation. Activation and clock start remain separate actions.

Take an encrypted backup and rehearse restoration into a new workspace. Verify intended concurrent load, storage and at least one full post-contest learning export. Record what was tested and any limitations before accepting the deployment.

Routine start and stop

Use the saved managed contest in the dashboard's server controls or an explicitly selected state. `manage.py start` starts only the dashboard. Native activation/startup does not start the contest clock. Pause the native contest and resolve pending judgements before planned maintenance. Use clean lifecycle shutdown; do not kill Java to bypass a refusal.

```
python -B lcpc/admin/manage.py status --state "$LCPC_STATE_DIR"
python -B lcpc/pc2ctl.py status
```

See the User Guide for event workflows. Read operational acceptance scripts before executing them; some submit test source, alter a dedicated trial or need retained local evidence. A passing source-level test suite is not proof that another laptop can reach your router.

Runtime environment variables

Variable	Purpose
LCPC_JAVA_HOME / JAVA_HOME	Java runtime home
LCPC_JAVA_BIN	Java executable override
LCPC_BUILD_JAVA_HOME	Bridge build JDK
LCPC_LIMA_BIN	Trusted limactl
LCPC_OPENSSL	Network OpenSSL
LCPC_PDF_PYTHON	PDF Python runtime
LCPC_STATE_DIR	Selected private contest state

Appendix B: Feature availability and support

A visible configuration is not an installed compiler, printer or external service. This matrix distinguishes working controls from read-only information, partial support and deployment work. It does not claim full parity with every Java desktop tool.

PC² Accounts & Permissions Available

Create/edit native identities, metadata and permission checkboxes; generate/import credentials. Creating a judge account does not start its service. Managed service accounts remain protected. Use individual accounts; start the deployed service through its lifecycle.

Languages & Compilers Partly supported

Manage language definitions; choose existing isolated C++17/C++20/Python profiles. Additional languages need a compiler/runtime and trusted sandbox profile. Saving a definition alone does not install them. Deploy and validate the language inside the isolated judge before enabling it.

Problem settings Available

Edit native problem identity, limits, flags, validator settings and balloon color. Only deployed isolated execution/validator policy may be active; test replacement does not rejudge. Use trusted deployment for custom executables.

Groups & Divisions Available

Create/edit group names, native IDs, scoreboard visibility and account membership. Group visibility controls do not replace private/public scoreboard freeze rules. Review membership and both scoreboard views before the event.

Clarification Categories Available

Create/edit active clarification categories used by the native contest. A category is a question-routing definition; it is not an announcement or an answer. Use Questions & announcements to send messages to teams.

Judgement Definitions Available

Create/edit verdict names, acronyms and active state. Native Accepted remains protected. A new verdict definition does not change a submission or add execution detection. Use the pending review queue for decisions and deployed judge support for detection.

Automatic Judge Clients

Requires external setup

Configure enabled judge accounts and native problem filters. The corresponding isolated judge process must already be deployed; settings do not create services. Provision the documented VM and start the judge lifecycle.

Contest & Team Display

Available

Edit native contest title, short name, team display, login policy and advanced flags. Configuration changes do not activate a separate contest or certify imported settings. Use Contests for independent events, and Clock & schedule for event timing.

Scoring Rules

Available

Read/edit the installed native scoring properties, including penalties and freeze behavior. These controls use the installed scoring implementation; they do not deploy arbitrary scoring plugins. Rehearse ties, pending approvals and public freeze before changing official rules.

Site Clocks

Available

Inspect/edit native site clocks and explicitly start/pause their configured scope. Changing elapsed time affects official event timing; server activation and clock start are separate. Review the site scope and schedule; pause before planned infrastructure changes.

Sites

Requires external setup

Edit permitted native sites and connection settings; protected local loopback remains fixed. Each remote site needs a separately configured PC² server and private credentials. Provision/test the remote site using native deployment procedures.

Verdict Notifications

Available

Configure per-problem final-verdict suppression and cutoff rules. Preliminary verdict delivery must remain suppressed by the judging workflow policy. Use announcements for general messages; approve held results from Submissions.

Balloon Settings

Requires external setup

Configure native balloon printing/mail settings. Printer drivers, devices, SMTP service and delivery need separate setup; no installation or test-mail guarantee. Use Problems for colors and Balloon delivery for acknowledgements; validate external delivery.

Balloon Delivery Available

List final accepted team/problem entries and confirm delivered or pending in one dialog. Acknowledgement records organizer intent, not proof of physical handover. Hidden diagnostic teams are excluded. Arrange physical balloon distribution separately.

Finalization & Medals Available

Edit native medal rank thresholds, certification state and comment. Certification does not resolve pending judgements or produce an institution-approved certificate design. Finalize decisions and verify standings before certifying; export the needed summaries.

Native Profiles Available

Edit existing native profile name, description and contest ID metadata. This is not a full desktop profile-switching interface or a clone of linked dashboard state. Use Contests to create, clone, activate and archive independent managed events.

Connections Available

Inspect native client/site connections and disconnect permitted non-runtime connections. Critical runtime clients are protected. Native connections are not a router device inventory. Use Team connections for observed browser presence; use router tools for network inventory.

Native Reports Partly supported

Generate supported native text reports with filters and owner reauthentication. Desktop file-only native report exporters need the desktop alternative. CSV/PDF summaries are not full recovery archives. Use the stated native desktop export or Exports & Backups as appropriate.

Submission Management Available

Filter submissions, adjust permitted elapsed times, recoverably remove/restore and release checkouts; request guarded rejudging. Rejudging can execute code and change official results. Bulk actions can partially complete after a connection failure. Inspect actual state before retrying; use submission details for final-verdict review.

Clarification Management Available

Inspect native questions, deletion/recovery and checkout state; send answers and announcements from the main page. Private/all-team audiences must be chosen deliberately; removal does not unsend a message already read. Check the recipient scope and retained question before confirming.

Private Judge Data Available

Inspect current private input/expected output and replace supported problem test pairs. Replacing current tests does not rejudge old attempts or reconstruct historical test revisions. Keep a separate data revision archive and request an explicit rejudge when justified.

Account Generation & Import Available

Preview and generate/import native accounts with group membership and credential export. Generated credentials are private; creating judge/site identities does not provision services. Use synthetic accounts for rehearsal and distribute real credentials privately.

Compare Source Files Available

Compare two retained source files without executing them. Text similarity is not proof of correctness, plagiarism or algorithmic equivalence. Inspect source and retained evidence with qualified human judgement.

Submit on Behalf of a Team Available

Owner can submit for an active team/problem while the clock runs. Creates an official attempt and can execute code through the isolated judge; requires reauthentication. Prefer the hidden diagnostic team for a rehearsal.

Contest Playback Available

Import validated native playback data and explicitly start/stop it. Only an empty technical contest; native playback starts clocks and continues after the browser closes. Use an independent rehearsal; never replay into a real event.

Event Feeds Read-only

Inspect retained/native Event Feed definitions and status. The installed upstream add/update handlers are unimplemented; this dashboard does not claim they work. Use supported native JSON/XML report exports, or contribute a tested upstream implementation.

Input Validation Results Read-only

Read saved native input-validation results. Arbitrary uploaded custom validators are not executed on the organizer host. Deploy trusted isolated validator support and validate it separately.

Shadow CCS Status Read-only

Inspect Shadow CCS state and existing imported-event information. External CCS credentials/connection are not provisioned here. Use the original native tools and a separately tested external CCS deployment.

Installed Java Plugins Read-only

List installed Java plugins and their native classes. No arbitrary JVM plugin installation, Swing windows or graphical print dialogs in the web dashboard. Use native PC² Administrator and trusted host deployment for supported plugins.

Service Logs Available

Read bounded/redacted tails of known service logs. Only supported files; log contents are not proof that every external service succeeded. Use trusted host diagnostics for deeper infrastructure inspection.

Reset & Recovery Available

Reset through the guarded native workflow. Requires full backup, exact event identity, owner reauthentication and explicit destructive confirmation. Prefer a new independent contest for a new event.

Feature Coverage Available

Show capability status and explicit limits in this section. This is an integration map, not parity with every upstream desktop dialog. Read the User Guide, Installation and Roadmap.

Organizer operating systems Not yet validated

Current deployment: macOS Apple Silicon with Lima VZ Linux ARM64 judge. This overlay is not certified on Windows/Linux organizer hosts or Intel Macs. Contribute isolated host support and run clean-machine acceptance.

One-click installer and fresh-host recovery Not yet validated

Engineer-assisted setup and consistent contest backups to a new workspace. No fully validated one-click clean-machine installer or full host/VM disk restore. Follow Installation and rehearse the whole deployment and recovery.

Historic test revisions and peak memory Unavailable

Current reference tests, retained native evidence and available execution times. No guaranteed historic dataset reconstruction or trusted peak memory metric. Use clearly labelled current-data learning replay; do not invent historical evidence.

Public cloud and multiple tenants Not yet validated

Private-LAN local organizer architecture with optional approved HTTPS admins. Not a certified public Internet service or multi-university tenancy layer. Separate threat model, deployment and independent security/load review.

Router device inventory Unavailable

Observed browser presence, verified session events and connection gaps. Cannot identify every router-connected device or prove uninterrupted physical network presence. Use router administration for hardware inventory with an appropriate privacy policy.

Team Learning Reports Available

Private per-active-team PDFs/ZIPs, source, statements and current-data isolated replay. Only deployed C++17/C++20/Python replay; expensive for large datasets and deliberate solution/test disclosure. Generate after the event, validate capacity and give each team only its own archive.

Appendix C: Ports, terms and attribution

Typical listeners

Service	Address / port	Exposure
PC²	127.0.0.1:50002	Local bridge / judge tunnel
WTI	127.0.0.1:8080	Behind team proxy
Team HTTP	SERVER-IP:8088	Approved private LAN
Team HTTPS	SERVER-IP:8443	Configured trusted TLS
Dashboard	127.0.0.1:8091	Organizer host only
Remote admin	SERVER-IP:8092	Optional HTTPS + user/IP approval

These are configuration examples, not real device credentials. Read actual access-center links. Never send a loopback dashboard address to teams. Public attachments and private judge tests are different data classes.

Terms that must stay distinct

Term	Meaning
Draft	Prepared event, not activated
Pending	No final verdict yet
Awaiting human	Automatic verdict is held for review
Freeze / Thaw	Hide / release public outcomes
First to Solve	Earliest final accepted submission for a problem
Report / Backup	Summary versus restorable state
Replay	Separately labelled analysis or technical playback

Attribution, sources and contribution

PC²: California State University, Sacramento; PC² Development Team: John Clevenger, Douglas Lane, Samir Ashoo, Troy Boudreau, and upstream contributors.

Dashboard development/customization: Boamtoal, in the context of Omar Al-Mukhtar University, Libya. Original system and contributor rights remain intact. PC², ICPC, university and contest logos belong to their owners; source licensing grants no trademark or endorsement rights.

Source: EPL-2.0. Nayuki QR retains its original MIT notice. Contributions to the first release are welcome. Never post private competition material in Issues. Use the source checkout matching this manual and review the published GitHub changes when updating.

Project source / Boamtoal
https://github.com/Boamtoal/pc2-university-platform

Original PC² project
https://github.com/pc2ccs/pc2v9

Original PC² website
https://pc2ccs.github.io/

License and attribution
https://github.com/Boamtoal/pc2-university-platform/blob/main/NOTICE.md

Appendix D: Other organizer systems - NOT TESTED

NOT TESTED WITH THIS PLATFORM. These are research-based starting points, not a working installer or a supported deployment. The exercised organizer host remains macOS on Apple Silicon. No Windows, Linux or Intel Mac acceptance run was performed. Documentation reviewed on 23 October 2026; upstream requirements may change.

This applies to the machine running the contest server. Teams using a browser on another OS do not need a server port. A remote administrator can use the approved HTTPS browser path to an existing supported server. PC²/WTI, this Python/Java overlay and the isolated execution service are separate components.

1. Understand what must be ported

The following list is based on this repository, rather than a claim from upstream documentation:

• Host discovery and executable paths: `lcpc/runtime_paths.py`, `lcpc/lan_info.py` and `lcpc/workspace_state.py`.

• Host processes, sockets, shutdown and restart: `lcpc/infra/vm.py`, `lcpc/infra/runtime_guard.py`, `lcpc/pc2ctl.py` and managed contest lifecycle.

• Guest architecture/image and bridge/tunnel provisioning: `lcpc/infra/judge-vm.yaml` and Linux judge scripts. Rebuild every architecture-dependent broker/binary for the chosen guest; update the image URL and its verified checksum together.

• LAN discovery, proxy, firewall, TLS and backup tools: `lcpc/infra/network.py`, `lcpc/infra/network_watch.py` and `lcpc/infra/secure_backup.py`. Environment overrides cover only some tools; audit remaining macOS paths.

• PDF fonts and dependencies: `docs/MANUALS.md`. A successful PDF build alone says nothing about judge isolation or scoring correctness.

Preserve separate organizer/guest boundaries and account permissions. An environment-variable change or a successful Java launch does not complete this work. Installing a language runtime must also happen inside the chosen isolated judge.

2. Linux organizer host: possible engineering route

Lima documents Linux hosts and a QEMU driver. This suggests a possible replacement for the current macOS VZ backend; it does not validate the existing template on Linux. Read Lima installation and VM driver selection.

Start on a separate lab machine with virtualization available. Inventory Java, Python, Node, OpenSSL, proxy and compiler dependencies for that distribution. Select a guest image matching the intended CPU and verify its provenance/checksum. QEMU's official installation page describes distribution packages; package availability is not a finished judge installation.

Port the host adapters and implement explicit service supervision, startup order, private state permissions and clean shutdown. On systemd distributions, service units are a possible supervision mechanism; see systemd.service. This repository provides no complete Linux organizer service set. Preserve the dedicated guest and trusted broker; do not replace them with direct student-code execution in the dashboard process.

3. Windows organizer host: possible engineering route

An engineer could investigate a Linux VM or WSL2 as a development environment. **Neither route has been tested for this platform.** WSL2 installation is not an isolated-judge security approval. Running a nested judge VM may require virtualization capabilities and additional architecture work.

Microsoft documents WSL installation and version checks. The following PowerShell commands prepare/inspect WSL only; they do not install or start this platform. The install command can require administrator rights and a restart:

```
wsl --install -d Ubuntu
wsl --list --verbose
```

See Microsoft WSL installation. Continue with the Linux porting tasks in a separate lab environment, including host adapters, trusted binaries and a genuinely separated execution boundary.

If service supervision is explored inside WSL, review Microsoft systemd guidance; enabling systemd does not by itself keep the WSL instance alive or make contest startup/recovery reliable.

LAN access requires its own design. Windows-local browser access does not establish access from team laptops. Review NAT/mirrored networking, Windows/guest firewall rules and intended listeners using Microsoft WSL networking. Do not forward the loopback administrator port to the LAN. Any future forwarding design must preserve trustworthy client identity for IP approvals instead of treating all forwarded clients as local administrators.

4. Intel Mac organizer host: possible engineering route

Lima supports multiple CPU/driver combinations, but this overlay currently pins an ARM64 judge profile. Build a separate x86_64 guest profile and matching trusted binaries; verify the chosen macOS/Lima driver requirements rather than merely renaming the ARM64 image. See Lima VM types and Lima architecture support.

Review executable locations, available macOS tools and PDF fonts. Measure compilation, execution and timeout behavior on that architecture. Passing on Apple Silicon does not certify Intel timing or capacity.

5. Isolation requirements that must survive the port

Ordinary VM examples may share host files automatically. Lima documents configurable host mounts; its defaults are not this platform's security policy. Explicitly review/remove host shares and credentials, and verify the final guest configuration against the intended boundary. See Lima filesystem mounts.

Preserve restricted execution users, filesystem/network restrictions, resource limits, output caps, timeout handling, cleanup and private judge data. Review the actual broker, guest kernel features and trust model. WSL, a container, systemd or a VM product name alone does not prove these properties.

6. Acceptance before claiming a supported port

Use synthetic accounts/problems and a separate event. Record OS/CPU, dependency versions, guest image hash, adapter changes and test results. At minimum verify startup/shutdown/reboot recovery; account roles/TOTP/IP approval; team login/upload from actual other laptops; accepted/WA/compile/runtime/timeout verdicts; held human review; ties/penalties/freeze/First to Solve; sandbox escape boundaries; PDF/replay exports; and backup restore. Rehearse sustained load and power/network interruption without official contest data.

Publish remaining failures and unsupported workflows with the proposed port. Open a focused pull request with sanitized fixtures and evidence. Until that work is reviewed, the compatibility table in Installation remains unchanged. Original PC² guidance is available from the official project; this porting sketch does not change upstream authorship or license terms.